MAT 3007 – Optimization

Algorithms for Unconstrained Optimization

Lecture 22

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Recap

Convex Optimization Problem

We call the optimization problems of the form:

- Minimize a convex function over a convex feasible region
- Maximize a concave function over a convex feasible region

convex optimization problems.

Otherwise, the problem is called a non-convex optimization problem.

- Convex problems are generally easier to solve efficiently.

Constraint Types

Lemma: Convex Level Sets

Let g be a convex function. Then, for any c, the level set $L_{\leq c} = \{\boldsymbol{x} : g(\boldsymbol{x}) \leq c\}$ is a convex set.

Lemma: Intersection of Convex Sets

The intersection of convex sets is a convex set.

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- Linear constraints $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and $\boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b}$, are convex constraints.
- Combinations of above constraints are convex constraints.

How Does Convexity Help?

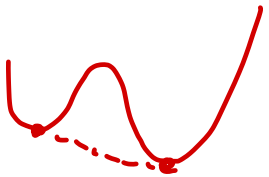
Theorem: Convexity and Global Solutions

Let $f : \Omega \to \mathbb{R}$ be a convex function and $\Omega \subseteq \mathbb{R}^n$ be a convex set. Then any local minimizer of the problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & \boldsymbol{x} \in \Omega \end{aligned}$$

is a global minimizer.

proof idea: something like



will not happen, because the line segment shold be above the graph for convex functions.

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is a global minimizer.

Proof: By contradiction, assume $\boldsymbol{x}^*$ is a local minimizer, and there exists $\bar{\boldsymbol{x}} \in \Omega$ such that $f(\bar{\boldsymbol{x}}) < f(\boldsymbol{x}^*)$. Then, using convexity,

$$f(\lambda \bar{\boldsymbol{x}} + (1-\lambda)\boldsymbol{x}^*) \leq \lambda f(\bar{\boldsymbol{x}}) + (1-\lambda) f(\boldsymbol{x}^*) < f(\boldsymbol{x}^*)$$

for any $0 < \lambda < 1$. This is a contradiction to: $\boldsymbol{x}^*$ is a local min. $\square$

Stationarity and Global Optimality: Unconstrained



Theorem: Stationarity & Global Optimality

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is convex on $\mathbb{R}^n$ and continuously differentiable at $\boldsymbol{x}^*$. Then, $\boldsymbol{x}^*$ is a global minimizer of $\min_{\boldsymbol{x} \in \mathbb{R}^n} f(\boldsymbol{x})$ if and only if $\nabla f(\boldsymbol{x}^*) = 0$.



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Remarks:

- In unconstrained convex optimization: stationarity (FONC) is sufficient and necessary for global optimality.
- Finding stationary points of a convex optimization problems providing global optimal solutions.

Proof: By the first-order characterization of convex function,

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}), \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,$$

when $\nabla f(\boldsymbol{x}^*) = 0$, we have

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The other side is shown by FONC for the unconstrained nonlinear optimization problems.

Stationarity and Global Optimality

Theorem: Stationarity & Global Optimality

Consider the constrained convex optimization problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\, i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\, j = 1, \ldots, p, \end{aligned}$$

where f and $\{g_i\}_{i=1}^m$ are convex, and $\{h_j\}_{j=1}^p$ are linear. Then any KKT point for this problem is a global minimizer.

Note : $\{x : g_i(x) \le 0 \ \forall i, h_j(x) = 0, \forall j\}$ is a convex constraint when g_i is convex and h_j is linear

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- Recall that in order to let the necessary direction (KKT conditions) to be true, we need CQ to hold.

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where f and $\{g_i\}_{i=1}^m$ are convex, and $\{h_j\}_{j=1}^p$ are linear. Then any KKT point for this problem is a global minimizer.

Remarks:

- If f and Ω are convex, then KKT conditions (FONC) are sufficient for global minimizers (no need of CQ at all).
- Recall that in order to let the necessary direction (KKT conditions) to be true, we need CQ to hold.
- If f is concave and g_i convex, h_j linear, then any KKT point of the problem $\max_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$ is a global maximizer of f.

Proof

Let $(\overline{\boldsymbol{x}}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}})$ be a KKT point.

1. Main Condition

$$\nabla f(\overline{\boldsymbol{x}}) + \sum_{i=1}^{m} \overline{\lambda}_i \nabla g_i(\overline{\boldsymbol{x}}) + \sum_{j=1}^{p} \overline{\mu}_j \nabla h_j(\overline{\boldsymbol{x}}) = 0.$$

2. Dual Feasibility

$$\overline{\lambda}_i \geq 0 \quad i = 1, ..., m.$$

3. Complementarity

$$\overline{\lambda}_i \cdot g_i(\overline{\boldsymbol{x}}) = 0 \quad \forall\ i = 1, ..., m.$$

4. Primal Feasibility

$$g_i(\overline{\boldsymbol{x}}) \leq 0, \quad h_j(\overline{\boldsymbol{x}}) = 0 \quad \forall\ i, \quad \forall\ j.$$

Proof

Because f and g_i are convex and h_j is linear, the Lagrangian $\boldsymbol{x} \mapsto L(\boldsymbol{x}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}})$ is convex. The main condition $\nabla_{\boldsymbol{x}} L(\overline{\boldsymbol{x}}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}}) = \mathbf{0}$ implies $\overline{\boldsymbol{x}}$ is a global minimum of $\boldsymbol{x} \mapsto L(\boldsymbol{x}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}})$ (by the theorem for unconstrained case). Thus,

$$
\begin{aligned}
f(\overline{\boldsymbol{x}}) &= f(\overline{\boldsymbol{x}}) + \sum_{i=1}^{m} \overline{\lambda}_i g_i(\overline{\boldsymbol{x}}) + \sum_{j=1}^{p} \overline{\mu}_j h_j(\overline{\boldsymbol{x}}) \quad \text{(complementarity)} \\
&= \min_{\boldsymbol{x}} \left\{ f(\boldsymbol{x}) + \sum_{i=1}^{m} \overline{\lambda}_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \overline{\mu}_j h_j(\boldsymbol{x}) \right\} \\
&\leq \min_{\{g_i(\boldsymbol{x}) \leq 0, h_j(\boldsymbol{x}) = 0, \forall i, j\}} \left\{ f(\boldsymbol{x}) + \sum_{i=1}^{m} \overline{\lambda}_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \overline{\mu}_j h_j(\boldsymbol{x}) \right\} \\
&\leq \min_{\{g_i(\boldsymbol{x}) \leq 0, h_j(\boldsymbol{x}) = 0, \forall i, j\}} f(\boldsymbol{x}).
\end{aligned}
$$

$$L(x, \bar{\lambda}, \bar{\mu}) = \underbrace{f(x)}_{\text{convex}} + \sum \bar{\lambda}_i \underbrace{g_i(x)}_{\text{convex}} + \sum \bar{\mu}_j \underbrace{h_j(x)}_{\text{linear}}$$

($\bar{\lambda}_i \geq 0$)

Convex in x

$\nabla_x L(\bar{x}, \bar{\lambda}, \bar{\mu}) = 0$

By the global optimality condition for unconstrained convex optimization, we have

$\bar{x}$ minimize $L(x, \bar{\lambda}, \bar{\mu})$.

Convex Optimization: Summary

- We have studied several elementary ways for identifying convexity of objective function.
- There is a sufficient condition for identifying convexity of constraint set.
- In convex optimization, solving to stationarity (FONC) or local solution means global optimizer.
- Software packages like CVX can be used to solve convex optimization.
- However, general nonconvex optimization problem is much more complicated.

Algorithms for Unconstrained Problems

Motivation for Nonlinear Optimization Algorithms



We now discuss how to solve nonlinear optimization problems.

- In some cases, the KKT conditions can be used to solve the optimization problem.
- However, those are ad hoc situations. In most situation, it is too complicated to directly find the KKT points from the KKT conditions, especially in high dimension.
- We want to have a robust procedure, an optimization algorithm), to solve the problems.

Unconstrained Problems

We start with the unconstrained problem:

$$\min_{\boldsymbol{x} \in \mathbb{R}^n} f(\boldsymbol{x})$$

We are going to study the following methods:

- Bisection method
- Gradient descent method
- Newton's method

General Process: Ideas

Optimization algorithms are typically iterative procedures that follow these steps:

- **Initialization**: Start from an initial point $\boldsymbol{x}^0$ and generate a sequence of iterates $\{\boldsymbol{x}^k\}$.
- **Termination**: The process stops when no further progress can be made or the current solution is deemed satisfactory.
- **Update Rule**: At each step, the new iterate $\boldsymbol{x}^{k+1}$ satisfy some criterion to improve the solution, e.g., $f(\boldsymbol{x}^{k+1}) < f(\boldsymbol{x}^k)$ (the objective value decreases).
- **Convergence**: Ideally, the sequence $\{\boldsymbol{x}^k\}$ converges to a local minimizer $\boldsymbol{x}^*$ (or even a global minimizer).

Recall that the simplex method follows the above paradigm.

Other convergence criterion: e.g. $f(x^k) \to f^*$

Definition: Convergence

Let $\{\boldsymbol{x}^k\}$ be a sequence of real vectors. Then $\{\boldsymbol{x}^k\}$ converges to $\boldsymbol{x}^*$ if and only if for every $\epsilon > 0$, there exists a positive integer K such that $\|\boldsymbol{x}^k - \boldsymbol{x}^*\| < \epsilon$ for all $k \geq K$.

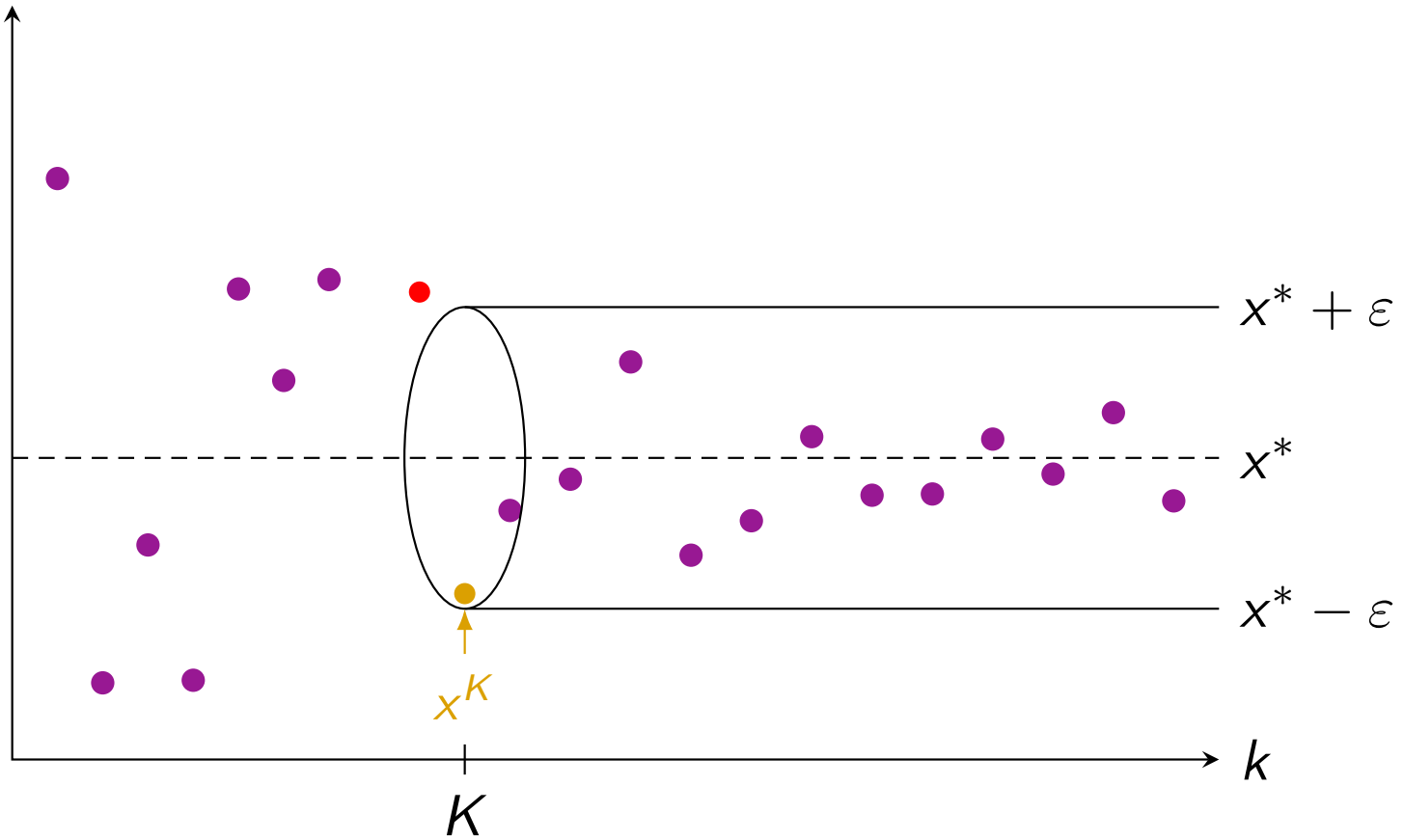
In all our discussions, we assume that $\|\cdot\|$ is the Euclidean norm:

$$\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x}^\top \boldsymbol{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

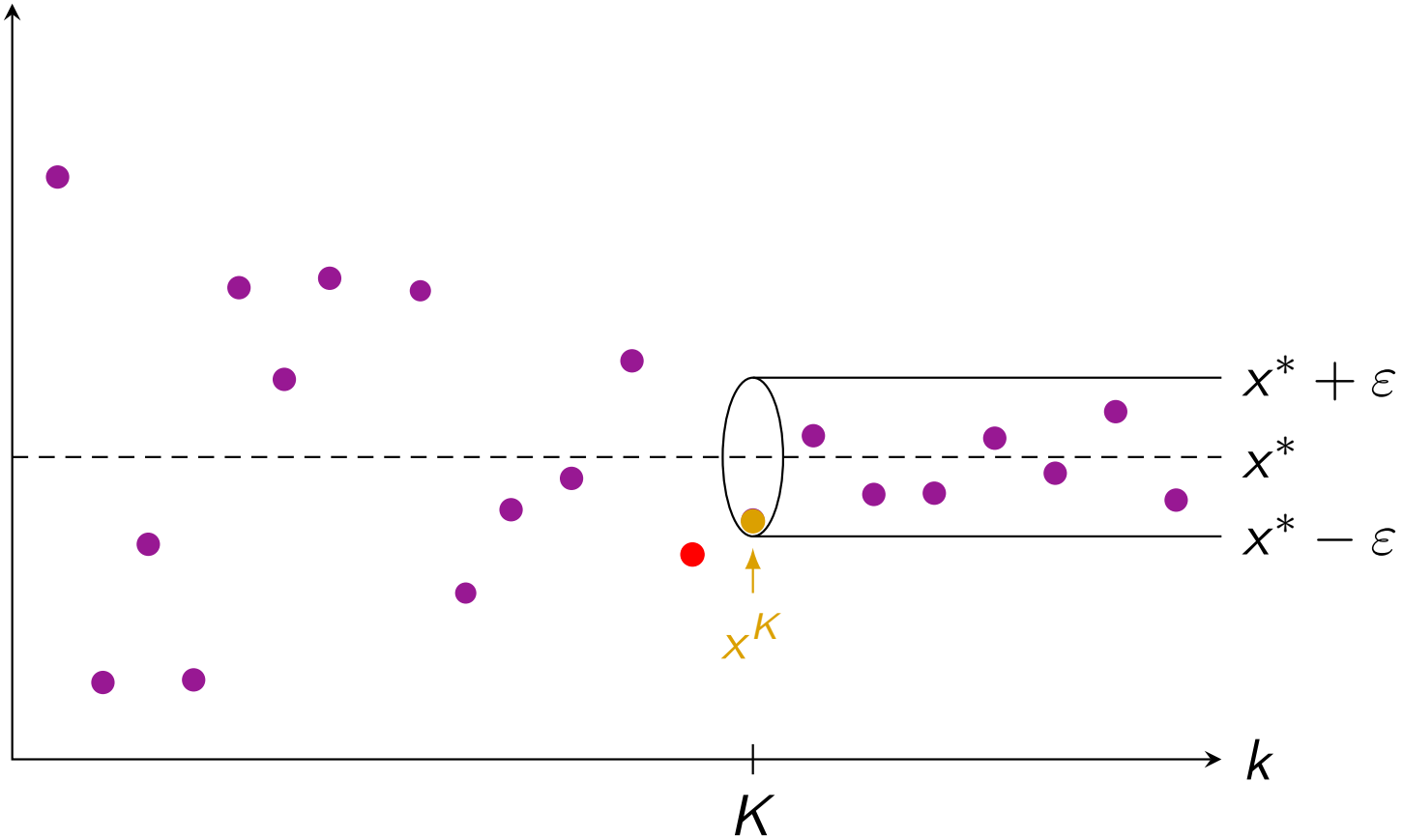
Examples of convergent sequences:

- $x^k := 1/k$ for all k; then $x^k \to 0$.
- $x^k := (1/2)^k$ for all k; then $x^k \to 0$.

Convergence: Illustration



Convergence: Illustration



Problems in $\mathbb{R}$

Single Variable Problem

Assume $f : \mathbb{R} \to \mathbb{R}$ is a single variable function.

Our Objective: find a local minimizer of f.

We introduce two methods:

- Bisection method.
- Golden section method.

Bisection Method

Bisection method uses the idea that the local minimizer must satisfy the first-order necessary conditions: $f'(x) = 0$.

Therefore, the problem becomes a root-finding problem for

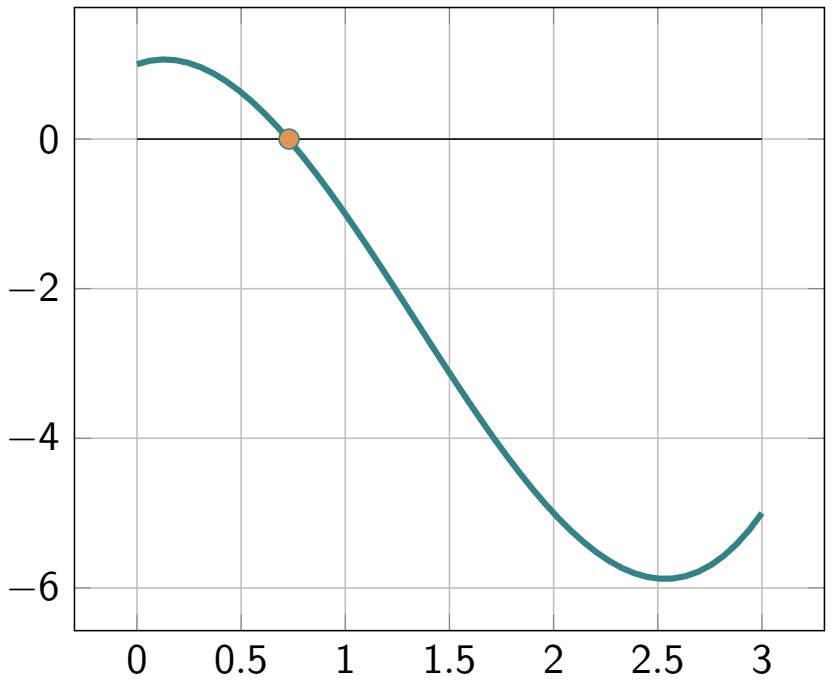
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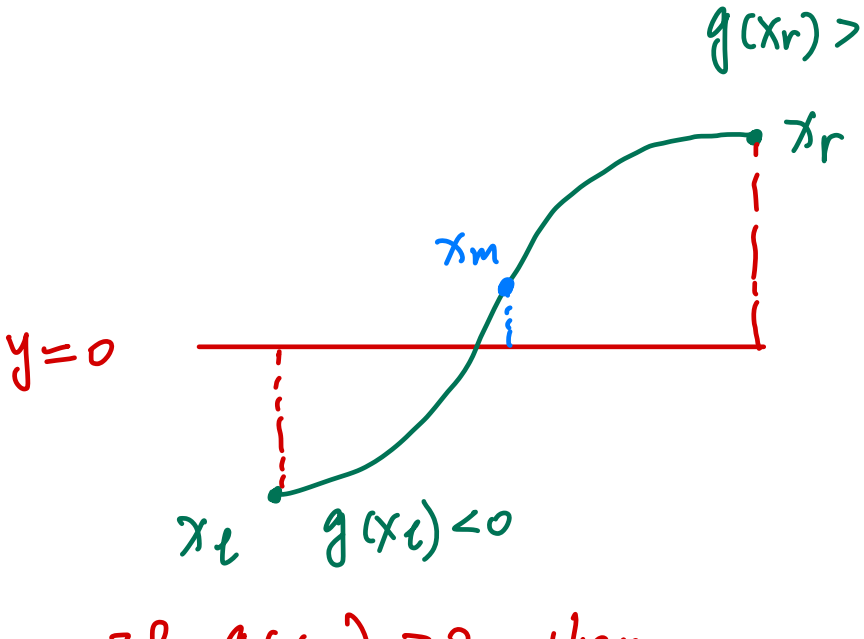
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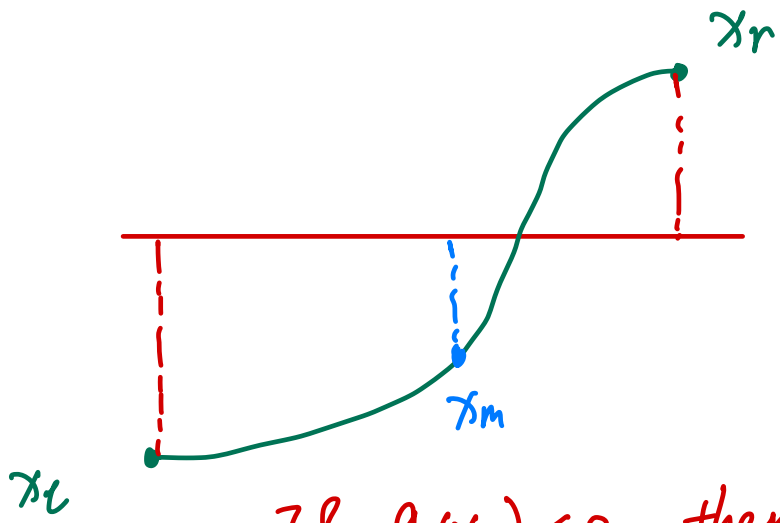
Root Finding Algorithm: Bisection Method

Assume we can find x_ℓ and x_r such that $g(x_\ell) < 0$ and $g(x_r) > 0$.

By the intermediate value theorem, if g is continuous, there must exist a root of g (i.e. x such that $g(x) = 0$) in $[x_\ell, x_r]$.



If $g(x_m) > 0$, then the root lies in (x_ℓ, x_m)



If $g(x_m) < 0$, then the root lies in (x_m, x_r)

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Bisection Method

1. Define $x_m = \frac{x_\ell + x_r}{2}$.
2. If $g(x_m) = 0$, then output x_m.
3. Otherwise:
   - If $g(x_m) > 0$, then let $x_r = x_m$. the root between (x_ℓ, x_m)
   - If $g(x_m) < 0$, then let $x_\ell = x_m$. the root between (x_m, x_r)
4. If $|x_r - x_\ell| < \varepsilon$: stop and output $\frac{x_\ell + x_r}{2}$, otherwise go back to step 1.

One can also set the stopping criterion based on $|g(x)| < \varepsilon$.

Bisection Method

In the bisection method, each iteration will divide the search interval to half. Therefore, to find an ε approximation of x^*, we need at most $\log_2(\frac{x_r - x_\ell}{\varepsilon})$ many iterations.

After k iterations, $x_r^k - x_\ell^k = (\frac{1}{2})^k (x_r - x_\ell)$.

If we want $x_r^k - x_\ell^k \le \varepsilon$, it's enough to pick $k = \log_2\left(\frac{x_r - x_\ell}{\varepsilon}\right)$.

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Applying the bisection method to f', we can find an approximate stationary point. If f is convex, this is an (approximate) global minimizer of f.

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Example: Use bisection method to maximize:

$$f(x) = \frac{xe^{-x}}{1+e^{-x}} \quad \rightsquigarrow \quad f'(x) = \frac{e^{-x}(1-x+e^{-x})}{(1+e^{-x})^2}$$

Bisection Method

```
function [x,gx] = bisection(g,xl,xr,options)

% Compute intial function values
gr  = g(xr); gl  = g(xl); sl = sign(gl);

if gl*gr > 0
    fprintf(1,'The input data not suitable!');
    x = []; gx = []; return
end

for i = 1:options.maxit
    xm  = (xl + xr)/2; gm  = g(xm);

    if abs(gm) < options.tol || abs(xl-xr) < options.tol
        x = xm; gx = gm; return
    end

    if gm > 0
        if sl < 0, xr = xm; else, xl = xm; end
    else
        if sl < 0, xl = xm; else, xr = xm; end
    end
end
```

Drawback of the bisection method: When solving (single variable, unconstrained) optimization problems, we require the knowledge (or computation) of f'.

Golden Section Method

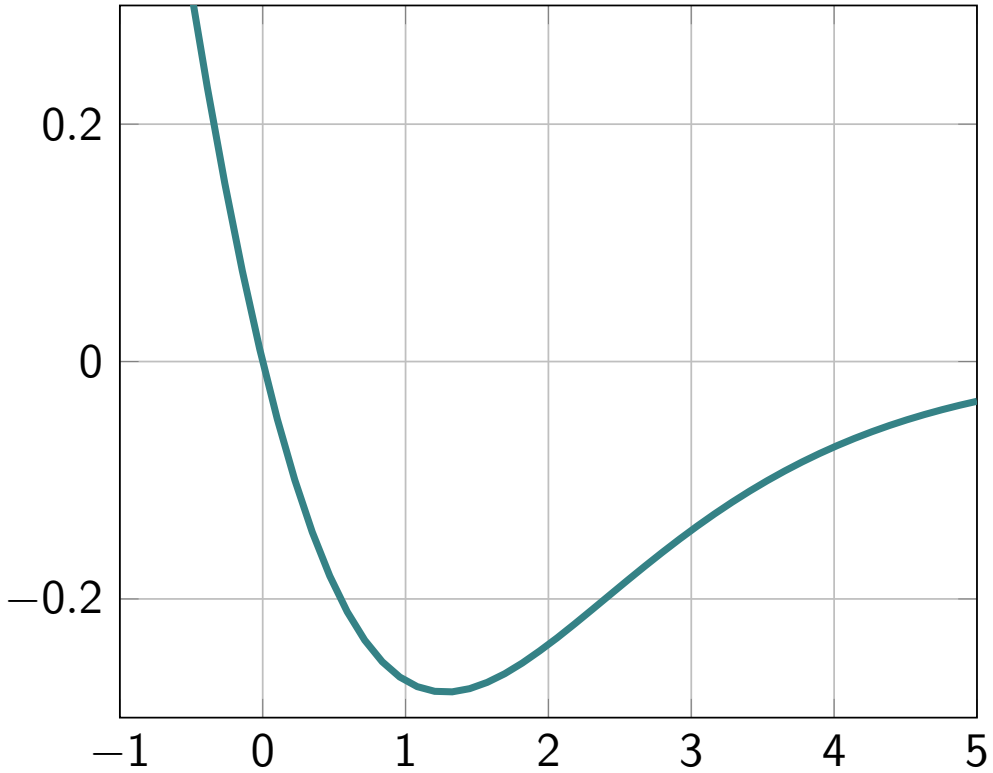
Drawback of the bisection method: When solving (single variable, unconstrained) optimization problems, we require the knowledge (or computation) of f'.

However, if we know that f has a unique local minimum x^* in the range $[x_\ell, x_r]$, then we still have a very efficient way to find x^*:

- We call f unimodal if it has a single local minimum or local maximum.
- Unimodal functions have the property that the local minimum is already global.

Example of a Unimodal Function

Consider $f(x) = -\frac{xe^{-x}}{1+e^{-x}}$:



This is a unimodal function, but not a concave function.

Golden Section Method

Assume we start with $[x_\ell, x_r]$. Assume $0 < \phi < 0.5$.

1. Set $x'_\ell = \phi x_r + (1-\phi)x_\ell$ and $x'_r = (1-\phi)x_r + \phi x_\ell$.
2. If $f(x'_\ell) < f(x'_r)$, then the minimizer must lie in $[x_\ell, x'_r]$, so set $x_r = x'_r$.
3. Otherwise, the minimizer must lie in $[x'_\ell, x_r]$, so set $x_\ell = x'_\ell$.
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$\rightsquigarrow$ This allows to save one function evaluation.

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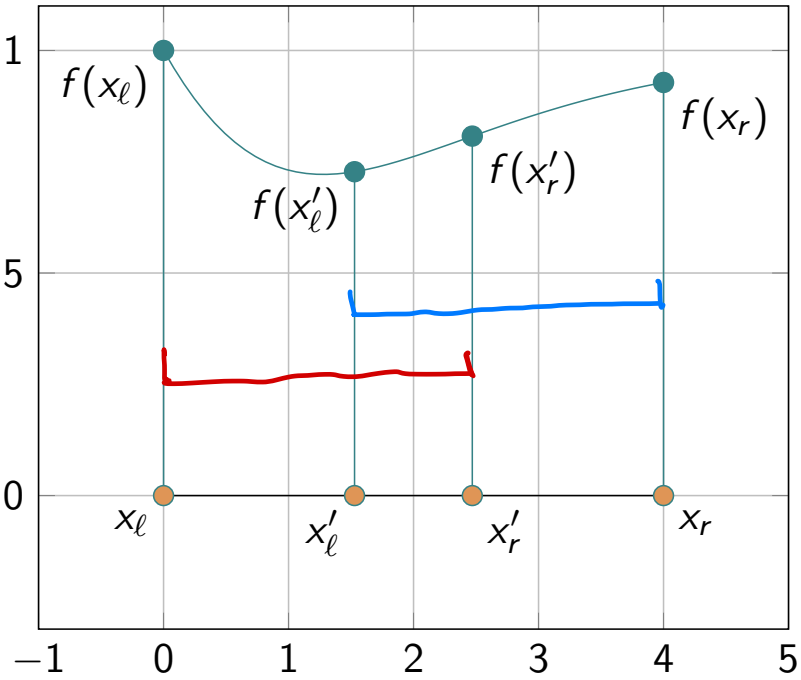
- This is true when

$$\phi = \frac{3-\sqrt{5}}{2} \quad \text{and} \quad 1-\phi = \frac{\sqrt{5}-1}{2} = 0.618.$$

Illustration and Example

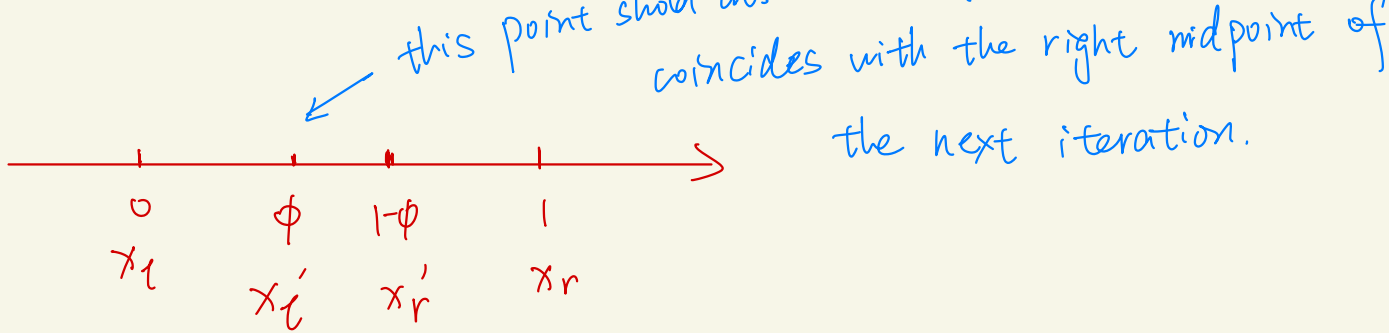
<1> if $f(x_\ell') < f(x_r')$ then the minimizer lies on the left segment.

<2> otherwise, the minimizer lies on the right segment.



Both the bisection and golden section method can be easily adapted for maximization problems. (Just adjust the comparison).

What is ϕ ?



$$x_l' = \phi = (1-\phi)^2$$

Solve ϕ : $\phi = 1 + \phi^2 - 2\phi$

$$\phi^2 - 3\phi + 1 = 0$$

Because $\phi \in (0, \frac{1}{2})$, we pick $\phi = \dfrac{3-\sqrt{5}}{2}$

Higher-Dimensional Problems

Higher Dimensional Problems

Consider the n-dimensional problem:

$$\text{minimize}_{\boldsymbol{x} \in \mathbb{R}^n} \quad f(\boldsymbol{x})$$

- There is no clear bisection or golden section in that case.

Solution and General Idea:

- Each time, we first find a search direction.
- Then, we search for a good next step along that direction. reduces to a one-dimensional problem).

General Framework for High Dimensional Search

Starting from the initial point $\boldsymbol{x}^0$, we generate a sequence of points:

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We call $\boldsymbol{d}^k$ the search direction (a vector) and α_k the step size (a scalar).

- The key is to choose a proper direction $\boldsymbol{d}^k$ at each iteration.
- $\boldsymbol{d}^k$ typically depends on $\boldsymbol{x}^k$.
- The step size α_k may be chosen in accordance with some line (one-dimensional) search rules (later).

Descent Directions

We assume that f is continuously differentiable.

Definition: Descent Direction

A vector $\boldsymbol{d} \in \mathbb{R}^n$ is a descent direction of f at $\boldsymbol{x}$ if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

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Important Observation:

- Taking a small enough step along a descent direction reduces the objective function value.

Descent Property

Let $\boldsymbol{d}$ be a descent direction of f at $\boldsymbol{x}$. Then, there is $\varepsilon > 0$ such that

$$f(\boldsymbol{x} + \alpha \boldsymbol{d}) < f(\boldsymbol{x}) \quad \forall\, \alpha \in (0, \varepsilon].$$

Abstract Descent Method: A First Scheme

Schematic Descent Directions Method

1. Initialization: Select an initial point $\boldsymbol{x}^0 \in \mathbb{R}^n$.

For $k = 0, 1, \ldots$:

2. If a stopping criterion holds, then STOP and $\boldsymbol{x}^k$ is the output.
3. Pick a descent direction $\boldsymbol{d}^k$.
4. Find a stepsize α_k satisfying $f(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k) < f(\boldsymbol{x}^k)$.
5. Set $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.

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Missing details:

- What is the initial point $\boldsymbol{x}^0$?
- How to choose the descent direction? What step size should be taken?
- What is the stopping criterion?

Gradient Descent and Stopping Criterion

Gradient Descent:

- One possible descent direction is $\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k)$:

$$\nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k = -\|\nabla f(\boldsymbol{x}^k)\|^2 < 0$$

  as long as $\nabla f(\boldsymbol{x}^k) \neq \boldsymbol{0}$.

- Choosing $\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k)$, the abstract descent method becomes the gradient descent method.

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Stopping Criterion:

- A popular stopping criterion is: $\|\nabla f(\boldsymbol{x}^k)\| \leq \varepsilon$ with tolerance $\varepsilon > 0$.

easy-to evaluate compared to $\|x - x^*\|$ as x^* is unknown.

$\rightsquigarrow$ We stop if $\boldsymbol{x}^k$ is an approximate stationary point.

$x^{k+1} = x^k - \alpha \nabla f(x^k)$ does not move if x^k is stationary.

Step Size Strategies

Step Sizes

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$$\alpha_k = \text{argmin}_{\alpha \geq 0} f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k). \tag{1}$$

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- If we compute the exact α_k in (1), this is referred to as the exact line search method for determining the step size.
- Solving for α_k is a one-dimensional optimization problem, and we can apply the methods discussed earlier to find its value.
- In some cases, we can even find the exact α analytically.

Example: $f(x) = x^2$ $\quad \nabla f(x) = 2x$

Gradient Descent with constant stepsize $\alpha_k = \alpha$.

$$x_{k+1} = x_k - \alpha(2x_k) = (1-2\alpha)x_k.$$

$$\Rightarrow \|x_{k+1} - x^*\|^2 = (1-2\alpha)^2 \|x_k - x^*\|^2 \quad \text{as } x^* = 0.$$

① if $\alpha > 1$, then $\|x_k - x^*\| \to \infty$

② if $\alpha = 1$, then $\|x_k - x^*\|$ stays constant with k.

③ if $\alpha \in (0,1)$, then $\|x_k - x^*\| \to 0$

Remark: Choice of stepsize is critical for convergence.

Example: Exact Line Search

Consider

$$f(\boldsymbol{x}) = \boldsymbol{b}^\top \boldsymbol{x} + \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \quad (\boldsymbol{A} \text{ positive definite}).$$

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To choose the step size, notice that we can explicitly compute

$$\begin{aligned} f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k) &= \boldsymbol{b}^\top(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k) + \frac{1}{2}(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k)^\top \boldsymbol{A}(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k) \\ &= \frac{1}{2}\alpha^2 (\boldsymbol{d}^k)^\top \boldsymbol{A}\boldsymbol{d}^k + \alpha(\boldsymbol{b}^\top \boldsymbol{d}^k + (\boldsymbol{x}^k)^\top \boldsymbol{A}\boldsymbol{d}^k) + f(\boldsymbol{x}^k) \end{aligned}$$

$(\boldsymbol{d}^k)^\top \boldsymbol{A}\boldsymbol{d}^k > 0$ as long as $\boldsymbol{d}^k \neq 0$

This is a quadratic function of α with positive second-order term.

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This is a quadratic function of α with positive second-order term. We can find the optimal $\alpha \geq 0$:

$$\alpha_k = -\frac{\nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k}{(\boldsymbol{d}^k)^\top \boldsymbol{A}\boldsymbol{d}^k} = \frac{(\boldsymbol{d}^k)^\top \boldsymbol{d}^k}{(\boldsymbol{d}^k)^\top \boldsymbol{A}\boldsymbol{d}^k}.$$

Today's lecture

- Bisecton and Golden Section Method
- Gradient Descent Method

Next lecture

- More Stepsize Options

Questions?